

Temporal Video Burst-Based Smart Attendance System Using Deep Face Recognition and Interval Voting

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Abstract

Traditional smart attendance systems which is image based often fail in real classroom environments due to occlusion, pose variation, lighting changes, and transient visibility. These limitations lead to unreliable recognition and inaccurate attendance records. To address this, we propose a Temporal Video Burst-Based Smart Attendance System that captures short video bursts (3 seconds) at fixed intervals during lectures. Each burst is decomposed into frames, where multi-face detection and recognition are performed using deep embeddings derived from a FaceNet-based architecture. A similarity-based matching mechanism compares detected embeddings with a pre-enrolled student database. To enhance robustness, a temporal presence score is computed per interval, and an interval voting strategy marks a student present only if recognized consistently across multiple intervals. This temporal aggregation significantly reduces false absence and false presence compared to single-frame methods. Expected outcomes include improved recognition reliability, reduced sensitivity to motion and lighting variations, and enhanced robustness in crowded scenarios. Future work will explore scalability to large lecture halls, multimodal fusion, edge deployment, and privacy-preserving mechanisms.

Keywords: Smart Attendance, Face Recognition, Temporal Aggregation, Video Burst, Interval Voting, Deep Learning, Classroom Automation